

EGAS

Promotion & Secondment Workflow System

Requirements & System Architecture Baseline

Version 5.2 - Implementation-Ready Design Baseline

19 August 2026

Item	Authoritative value
Status	Implementation-ready requirements, architecture and detailed design baseline for the new repository. Business behavior remains aligned with v5.1; v5.2 adds normative ERDs, component/state/sequence/security/deployment diagrams, physical schema constraints and coding-agent work packages.
Repository strategy	Create a new repository and clean database baseline. The previous repository remains an archival reference; its local migrations and workflow schema are not carried forward as mandatory history.
Core redesign	Department/unit hierarchy with one active manager and subordinate employees; manager-controlled assignment/reassignment; employee-to-manager review loop; manager-only external routing/signing.
Business workflows	Promotion: HR -> ORG -> HR -> AUTH -> HR for SAME_POSITION; if any candidate is promoted to OTHER_POSITION within the same department/routing unit, the request follows HR -> ORG -> HR -> AUTH -> P4O ORG confirmation -> HR. Secondment: HR -> ORG -> AUTH -> ORG -> HR.
Official signers	HR manager, ORG manager, and the applicable AUTH manager/deputy. Later HR/ORG review stages do not create duplicate signatures.
Data integration	Approved annual XLSX remains the authoritative operational employee snapshot; new experience/job-tenure and department data are normalized into the annual snapshot.
Document baseline	New Promotion official form contract is treated as PDF template V3.0. Final immutable official PDF is created when all required fields/approvals are complete.

1. Document control, precedence and redesign decisions

1.1 Revision control

Version	Date	Status	Description
4.0	19 Aug 2026	Historical implementation-aligned baseline	Flat operational roles, shared Organization claim model, originating-HR ownership, P4O cross-routing branch, two official signer blocks plus existing signoff behavior.
5.0	19 Aug 2026	Superseded stakeholder redesign draft	Introduced manager/employee hierarchies and the clean-repository redesign, but incorrectly removed P4O after interpreting the same-department position rule as eliminating the conditional Organization confirmation stage.
5.1	19 Aug 2026	Historical corrected stakeholder redesign baseline	Restored conditional P4O using the same-department OTHER_POSITION trigger and locked hierarchy/signature/business behavior.
5.2	19 Aug 2026	Current implementation-ready design baseline	No business-rule change from v5.1. Adds detailed implementation blueprints: repository reuse policy, module/component architecture, StageExecution/request state machines, authorization flow, domain ERDs, transaction boundaries, runtime sequences, deployment trust boundaries and SDLC/agent dependency map.

1.2 Authoritative precedence

Priority	Source	Rule
1	This v5.2 baseline	Current business/workflow/security/data/architecture authority for the new repository.
2	Latest stakeholder clarifications and approved official forms	Used to resolve any ambiguity or later approved change.
3	New repository implementation and immutable migrations	Implementation authority after the new baseline is established; code must not silently override business intent.
4	v4.0 baseline / prior repository	Historical traceability and reusable technical patterns only where they do not conflict with v5.2.

1.3 Major v4.0 rules superseded by v5.2

v4.0 rule	v5.2 disposition	Current rule
Flat Employee Affairs / Organization / Approving Authority roles	SUPERSEDED	Operational users belong to one operational unit and are either MANAGER or EMPLOYEE. Work authority is hierarchy- and assignment-based.
One account may hold multiple operational roles with active-role switching	REMOVED	Operational accounts belong to one operational unit/hierarchy. Admin accounts are administrative-only. No operational permission union.
Organization shared unassigned claim queue	SUPERSEDED	A unit manager receives incoming work and explicitly assigns it to a subordinate employee or takes it personally.
Originating HR user owns later returns/restarts/recalls	SUPERSEDED	HR manager owns reject recovery, restart/cancel authority, and later HR-stage assignment decisions.
P4O Organization confirmation stage	RETAINED - TRIGGER CORRECTED	Cross-department/routing-unit targets remain prohibited. P4O is retained as a conditional Organization confirmation stage when AUTH selects OTHER_POSITION for any Promotion candidate, meaning a different position within the same department/routing unit. If every candidate remains SAME_POSITION, P4 routes directly to P5.
Official signer blocks are HR + Organization only	SUPERSEDED	Official form carries HR manager, ORG manager and applicable AUTH manager/deputy signatures.
Received-stage immutable PDF per actor/claim	SIMPLIFIED	Workflow uses an on-demand current PDF view; one immutable final official PDF is frozen at completion. Audit/history preserves who received/edited/assigned/signed.
Migrations 001-007 are mandatory implementation history	NOT CARRIED INTO NEW REPO	All prior work was local/disposable; new repository starts with a clean migration 001 representing the v5 physical schema.

2. Purpose, scope and terminology

2.1 Product objective

- Digitize EGAS Promotion (ترقية) and Secondment (ندب) workflows with explicit department/unit hierarchy and manager accountability.
- Separate preparatory work by employees from manager authority to approve, sign, reassign and route work outside the unit.
- Use the approved annual employee XLSX as the authoritative operational population; normal runtime has no direct SAP HCM/database integration.
- Provide Arabic-first RTL workflow screens, manager inboxes, employee assignments, notes/history, in-app notifications, audit evidence, on-demand form preview and immutable final official PDFs.
- Preserve immutable audit evidence, stage executions, signoffs and completed documents while allowing controlled correction loops and whole-request restart after rejection.

2.2 Core terminology

Term	Meaning
شئون العاملين / HR	Operational HR unit. Its manager controls initiation/final HR review, assignment, signing at the first HR stage, and reject recovery.
إدارة التنظيم / ORG	Organization unit. Its manager assigns preparation/review work, signs the first ORG stage, and later performs non-signing review where the workflow returns to ORG.
سلطة الاعتماد / AUTH	Approving Authority hierarchy for a configured department/routing unit (نيابة/مساعد). Each AUTH unit has one active manager/deputy and subordinate employees.
Operational Unit	A hierarchy container with one active manager and zero or more employees. Active v5 configuration includes HR, ORG and one AUTH unit per routing authority/routing unit.
Manager	The only hierarchy member who can assign/reassign work, approve employee work, sign when required, and route to the next business stage.
Employee	A subordinate operational worker who may edit assigned stage data and submit it to the manager, but cannot sign or route outside the unit.
Business Stage	A cross-unit workflow step such as P2 Organization Preparation or S3 AUTH Decision.
Stage Execution	One immutable execution/attempt of a business stage. A return to a previously signed stage creates a fresh execution instead of rewriting the previous signoff.
Work Assignment	Current manager-to-employee assignment inside one Stage Execution.
Internal Correction	Manager returns work to the current employee inside the same business stage before final manager acceptance/signoff.
Inter-stage Correction	The current unit returns the request to the immediately previous business-stage manager for correction. Prior evidence remains immutable.
Reject	Complete rejection routed to the HR manager. HR manager chooses whole-request restart as a new iteration or final cancellation.
Iteration	One immutable whole-request pass. Restart after rejection

	creates a new iteration from P1/S1.
--	-------------------------------------

2.3 In scope

- Local application accounts and Admin-managed operational hierarchy.
- Annual XLSX staging, validation, normalization, routing/department mapping and activation.
- Manager inbox, assignment/reassignment, manager self-work, employee work queues and manager review.
- Promotion and Secondment business flows, internal/inter-stage correction, rejection, restart and cancellation.
- Manager-only signatures with fresh current-password reauthentication.
- Three-signer Promotion official form V3; Secondment official form behavior with the same hierarchy/signoff principles.
- On-demand current form preview, immutable final official PDF, separate audit/report evidence.
- Arabic-first desktop/laptop web application, role/unit-scoped notifications and history.

2.4 Out of scope / deferred

- Direct SAP HCM/database access, write-back, OData/RFC/BAPI, SAP BTP/Fiori runtime dependency.
- Public registration, employee self-service, public Internet access, mobile application and email notifications unless separately approved.
- Automatic vacancy sourcing or independent HR eligibility/calculation engines.
- Advanced BI analytics beyond operational administration/readiness.
- Production infrastructure specifics, penetration testing and final corporate cybersecurity approval until EGAS IT defines them.

3. System architecture baseline

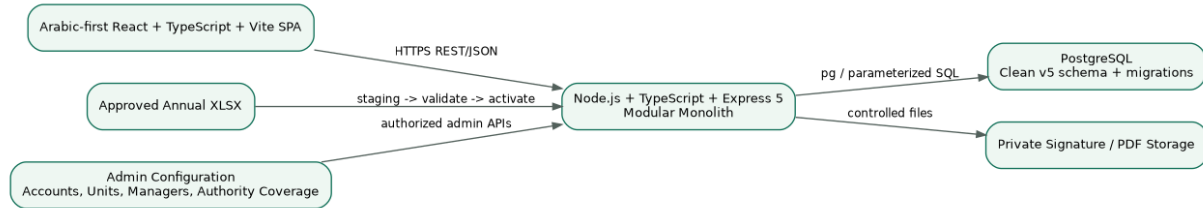


Figure 1 - v5.2 high-level architecture

3.1 Architecture freeze

Target stack

React + TypeScript + Vite (single Arabic-first SPA) -> same-origin HTTPS REST/JSON -> Node.js + TypeScript + Express 5 modular monolith -> pg / parameterized SQL -> PostgreSQL.

- Single repository for frontend/backend/shared contracts and documentation.
- No microservices requirement for v5 MVP; modular monolith boundaries are used to keep workflow, identity, data import, audit and PDF concerns separate.
- Normal browser operation never connects directly to PostgreSQL and never consumes the annual Excel workbook directly.
- Annual XLSX follows controlled staging -> validation/revalidation -> explicit activation -> immutable annual snapshot.
- Signature assets and generated final PDFs live in private controlled storage; only authorized backend routes serve them.

3.2 Suggested repository structure

```

apps/
  web/           React + TypeScript + Vite
services/
  api/
    src/
      modules/
        auth/
        admin/
        hierarchy/
        employee-data/
        workflow/
        signatures/
        documents/
        notifications/
        audit/
      db/migrations/
      middleware/
      shared/
  docs/
    requirements/
    architecture/
    api/
  
```

3.3 New-repository / migration strategy

- Archive the v4 repository and tag its last working state. Do not delete it; it remains implementation reference and historical evidence.
- Create a new v5 repository from a clean application skeleton or carefully copied reusable infrastructure modules.

- Create `001\_initial\_v5\_schema.sql` representing the new hierarchy/workflow model directly. Do not copy old migrations 001-007 as mandatory history.
- Reuse proven security/session/import/PDF patterns where they still fit, but review copied code against v5 authorization and workflow rules before enabling it.
- Never seed synthetic operational staff/authority data into a real pilot database merely to satisfy readiness checks.

4. Accounts, operational hierarchy and authorization

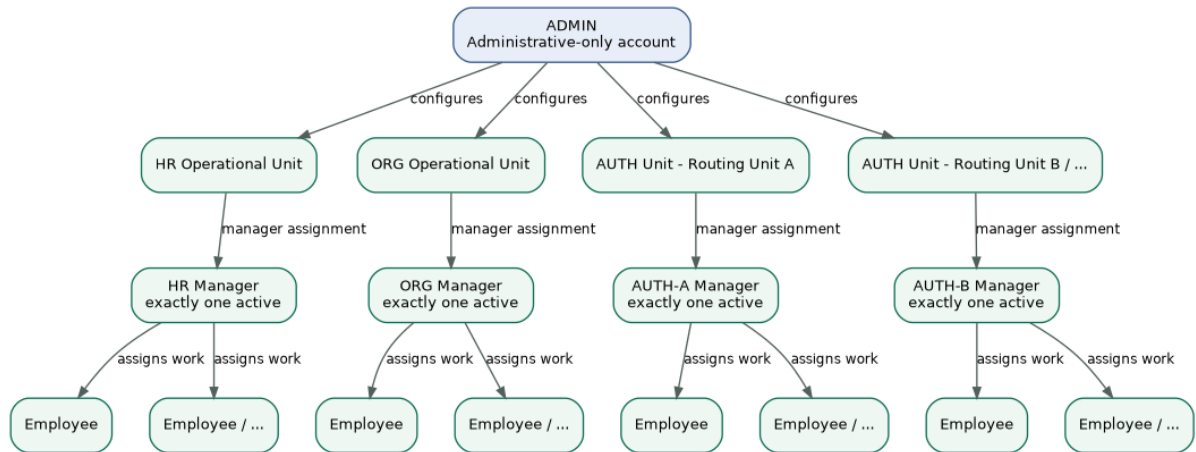


Figure 2 - Operational hierarchy model

4.1 Account model

Account type	Membership rule	Capabilities
ADMIN	Administrative-only; no operational unit membership required.	Manage accounts, operational units, managers/employees, manager replacement/delegation, routing/authority configuration, annual-data administration, audit/reporting. Cannot perform operational workflow actions merely because it is Admin.
OPERATIONAL - MANAGER	Exactly one active operational-unit membership.	All unit-stage edit capability plus assignment/reassignment, manager review, signature where required, cross-stage routing, reject/correction actions permitted by workflow.
OPERATIONAL - EMPLOYEE	Exactly one active operational-unit membership.	View/edit only the currently assigned work; submit to manager; append authorized notes. Cannot assign, sign or route externally.

Simplification decision

v5 intentionally removes operational multi-role accounts and active-role switching. This reduces authorization complexity and prevents accidental permission union. A user who changes organizational responsibility should be changed through controlled Admin membership/assignment history rather than simultaneous operational roles.

4.2 Operational units and manager rules

- Each operational unit has exactly one active manager assignment at a time.
- HR and ORG each have their configured unit manager and subordinate employees.
- Each configured AUTH routing authority / نيابة has its own AUTH operational unit, one active manager/deputy and subordinate employees.
- Admin can replace/reassign a manager. Manager assignment history is effective-dated/immutable so old signoffs continue to identify the real signer at the time.
- An acting/replacement manager becomes the active manager for new actions; old manager evidence is never rewritten.

4.3 Request visibility

Actor	Visibility / edit rule
Current unit manager	Can view the active stage for the unit, its history, current assignee and permitted data. Can edit directly or assign/reassign.
Current assigned employee	Can view and edit only the currently assigned stage work within server-authorized fields/actions.
Former assignee	Loses active edit access immediately after reassignment. Historical actions/notes remain in audit/history.
Other employee in same unit	No access merely by unit membership unless assigned.
Previous/future unit	Access only when workflow/history rules authorize the specific request/stage; no global operational browsing.
Admin	Administrative/audit access only according to Admin permission. Admin is not a workflow actor.

4.4 Reassignment rules

- When a request returns to a unit later in the workflow, the employee who previously worked on the corresponding request/stage is shown as the default suggested assignee.
- The manager may accept the default, choose another subordinate, or take the work personally.
- Reassignment is audited with from-assignee, to-assignee, manager, stage execution and server timestamp.
- Only the current manager may reassign; employees cannot delegate their own work.

5. Generic stage-processing model



Figure 3 - Manager/employee processing inside a business stage

5.1 Internal stage lifecycle

State / action	Rule
Manager Inbox	Incoming business stage is addressed to the responsible operational-unit manager, not directly to a random employee.
Assign	Manager selects a subordinate employee. If the request previously visited this unit, the prior worker is the default suggestion but is not mandatory.
Take it myself	Manager may work the stage directly without assigning an employee.
Employee In Progress	Only the current employee assignee and manager can edit authorized stage fields.
Submit to Manager	Employee submits completed stage work to manager review. This does not route to the next department.
Internal Correction	Manager sends the work back to the current employee with a required reason. The business stage does not change.
Manager Review	Manager may correct the work directly, reassign it, return it internally, perform inter-stage correction/reject if permitted, or approve/sign.
Manager Signoff	Only signing stages require signature asset + current-password reauthentication. Signoff is immutable.
Advance	Only the manager can route the stage to the next business stage.

5.2 Stage execution and signed-data immutability

Critical rule

A previous manager signoff is never edited. If a later stage returns for correction to a stage that already produced a signature, the system creates a new Stage Execution for that business stage. The old execution/signoff remains historical evidence; the corrected execution must receive a fresh manager signoff before advancing again.

- Internal manager-to-employee correction before manager acceptance/signing stays within the same Stage Execution.
- Inter-stage correction creates a new execution of the destination business stage when needed; it does not erase the earlier execution.
- The current request view uses the latest accepted execution for each stage while history/audit can display all executions.

6. Correction, rejection, restart and cancellation

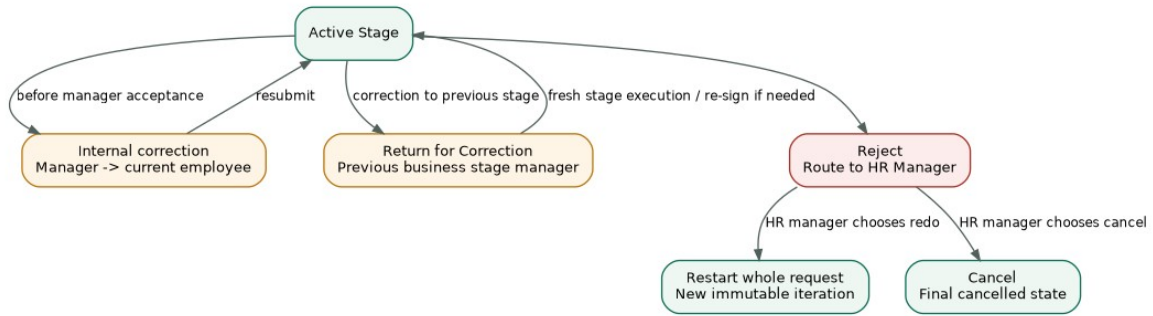


Figure 4 - Exception control model

6.1 Three distinct actions

Action	Destination	Effect
Internal Correction	Current stage -> current assigned employee	Manager requests correction without leaving the business stage. Reason required. Employee corrects and resubmits to the same manager.
Return for Correction	Current stage -> immediately previous business-stage manager	Request moves one business stage backward for correction. Reason required. Prior stage evidence remains immutable; a new stage execution is created where appropriate.
Reject	Any permitted rejecting stage -> HR manager	Complete request rejection. Current iteration closes as rejected/returned-to-HR-control. HR manager decides either whole-request restart or cancellation.

6.2 Reject recovery

- Reject never routes to an ordinary HR employee. It routes to the HR manager.
- HR manager sees the rejection reason and complete prior iteration history.
- `Restart Whole Request` creates the next immutable WorkflowIteration and begins again at P1/S1. Prior candidates/evidence may be copied only through an explicit restart preparation function; prior evidence is not mutated.
- `Cancel` sets a final CANCELLED status. Normal operational users cannot reopen a cancelled request.

6.3 Return-to-previous-stage examples

Current stage	Return for Correction destination
P2 ORG	P1 HR manager
P3 HR	P2 ORG manager
P4 AUTH	P3 HR manager
P4O ORG	P4 AUTH manager
P5 HR	P4O ORG manager when P4O occurred; otherwise P4 AUTH manager
S2 ORG	S1 HR manager

S3 AUTH	S2 ORG manager
S4 ORG	S3 AUTH manager
S5 HR	S4 ORG manager

Conditional P4O retained in active v5.2

P4O is not a cross-department routing stage. Promotion candidates must remain within the same department/routing unit. If every P4 decision is SAME_POSITION, P4 advances directly to P5. If any candidate is OTHER_POSITION - a different position within that same department/routing unit - the whole request advances to P4O for Organization confirmation before P5.

7. Promotion workflow

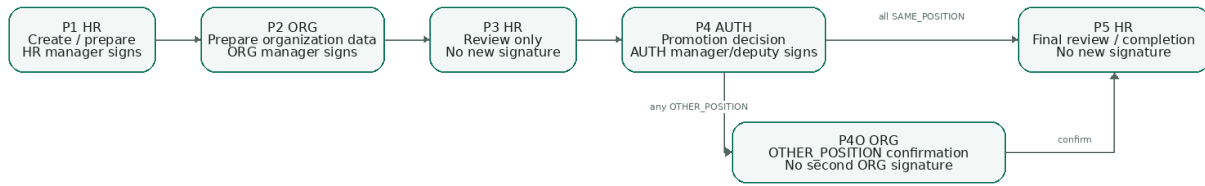


Figure 5 - Promotion business workflow

7.1 Promotion stages

Stage	Responsible unit	Business responsibility	Signature / transition
P1	HR	Create request; add candidates from active annual snapshot; complete HR-owned fields; prepare notes/form data. Employee may prepare; HR manager reviews.	HR manager signature required with fresh password -> P2.
P2	ORG	Complete Organization-owned candidate/form data and required organizational preparation.	ORG manager signature required with fresh password -> P3.
P3	HR	Review Organization preparation. Manager assigns/reviews as needed.	Review only; no new HR signature. Approve -> P4; return/reject per rules.
P4	AUTH unit for request routing	AUTH employee may prepare recommendation/decision; AUTH manager/deputy reviews and finalizes Promotion decision. SAME_POSITION keeps the candidate on the current position. OTHER_POSITION selects a different position within the same department/routing unit; cross-department/routing-unit targets are rejected.	AUTH manager/deputy signature required with fresh password. If every candidate is SAME_POSITION -> P5. If any candidate is OTHER_POSITION -> P4O.
P4O	ORG	Conditional Organization confirmation for Promotion OTHER_POSITION decisions. The ORG manager receives the stage and may assign it to the previous ORG employee by default or another subordinate. AUTH signed decisions are read-only; Organization confirms the within-department placement and may add permitted Organization-owned information/notes.	No second ORG signature. Confirm -> P5. Internal correction may return to the assigned ORG employee; inter-stage correction -> P4 AUTH; Reject -> HR manager for restart/cancel decision.
P5	HR	Final HR review and completion.	No new HR signature. Completion freezes final official PDF V3.

7.2 Promotion decision rules

- Per candidate, Promotion position decision remains `SAME\_POSITION` or `OTHER\_POSITION`.
- `OTHER\_POSITION` must identify a different proposed/nominated job within the same department/routing unit as the candidate/request. Cross-department/routing-unit targets are rejected by backend validation.
- The official form also carries `التوصية` with the approved form values such as `ترشيح / تأجيل`; this is captured as an AUTH-controlled candidate decision field and does not authorize cross-department routing.
- The official `الوظيفة المرشح لشغلها` field displays the effective nominated job. For SAME_POSITION this resolves to the current job; for OTHER_POSITION it displays the selected/entered different position in the same department/routing unit.
- P40 trigger: if any candidate has an OTHER_POSITION decision, the whole Promotion request enters P40 for Organization confirmation before final HR review. If all candidates are SAME_POSITION, P4 bypasses P40 and advances directly to P5.
- Submitted AUTH manager decision/signoff is immutable within its Stage Execution. Correction requires the controlled return/new-execution mechanism.

7.3 Promotion validation

- All candidates must belong to the active annual snapshot and resolve to the request routing unit.
- Required new official-form data (department/general administration, current-job start/tenure, experience values and performance report where defined) must be available before final completion.
- Manager cannot sign while mandatory stage-owned fields are incomplete.
- P5 cannot complete unless current accepted P1, P2 and P4 signoffs exist and all required final form fields are complete.

8. Secondment workflow

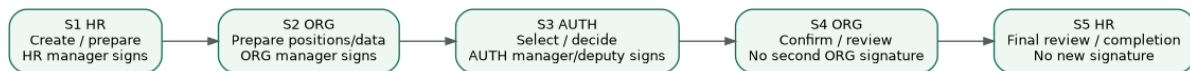


Figure 6 - Secondment business workflow

8.1 Secondment stages

Stage	Responsible unit	Business responsibility	Signature / transition
S1	HR	Create request; add candidates and complete HR-owned data.	HR manager signature required with fresh password -> S2.
S2	ORG	Prepare required secondment data including proposed positions, organizational dependency, qualification/compliance status and any approved Organization-owned fields.	ORG manager signature required with fresh password -> S3.
S3	AUTH unit for request routing	AUTH employee may prepare; AUTH manager/deputy selects/approves the position/decision. Target position must remain within the same department/routing unit.	AUTH manager/deputy signature required with fresh password -> S4.
S4	ORG	Review/confirm the AUTH-selected result without silently replacing the signed authority decision.	No second ORG signature; approve -> S5 or return/reject per rules.
S5	HR	Final HR review and	No new HR signature.

		completion.	Completion freezes final official PDF.
--	--	-------------	--

8.2 Secondment decision constraints

- Secondment positions are prepared/maintained under Organization-controlled data and AUTH selects/approves according to the business stage.
- Target positions must belong to the same department/routing unit; no cross-department/routing-unit secondment target is allowed in v5.
- S4 Organization cannot silently replace the AUTH manager/deputy signed selection. A required change must use Return for Correction so the authority evidence is preserved.
- The ORG manager signature created at S2 remains the official Organization signature. S4 is a review stage and does not create another ORG signer block.

9. Official forms, signatures and PDF lifecycle

9.1 PDF lifecycle simplification

Document state	Rule
CURRENT / DRAFT VIEW	On-demand authorized rendering of the current request for workflow users. It is not immutable evidence and may change as authorized edits occur.
FINAL	One immutable official business PDF created/frozen when the final HR stage completes and all required fields/signoffs exist.
AUDIT LOG	Separate printable audit/history report. It is not part of the official business form.

- v5 does not create a separate immutable received-stage PDF for every manager/employee handoff. Assignment and received/edit history is preserved in StageExecution/WorkAssignment/AuditEvent records instead.
- PDF generation failure must never mutate or advance workflow state.
- Final PDF source data is frozen and checksum-verifiable; later annual imports, account changes, manager replacement or template activation do not rewrite historical completed documents.

9.2 Signature model

Signer	Promotion	Secondment	Rule
HR manager	P1	S1	First HR stage only. Later HR review/final stages do not create a replacement HR signature.
ORG manager	P2	S2	First Organization stage only. S4 later Organization review does not create a second ORG signature.
AUTH manager / deputy	P4	S3	Applicable manager/deputy for the request routing authority. New third official signer.

- Immediately before every actual signature, backend verifies the raw current account password of the authenticated manager/deputy.
- Missing/wrong password creates no signoff. Wrong attempts produce safe security evidence but never persist the password.
- Signoff freezes signer account identity, operational unit, manager assignment, display name, job title, signature asset hash and server sign time.
- Signature blocks on the official form appear on one horizontal signer area consistent with the stakeholder-supplied layout; the AUTH/deputy signer occupies the additional left-most signer position in the new Promotion form.

9.3 Promotion official form V3 contract

Template version

New Promotion snapshots use `EGAS-OFFICIAL-PROMOTION-AR-3.0`. The v2 form belongs to the historical repository/baseline. The new repository does not need to render old v2 data unless historical migration is later explicitly requested.

Arabic field / area	Source / rule
م	Serial number within the applicable official form grouping/section.
رقم العامل	Frozen Personnel Number from annual employee snapshot.
اسم المرشح	Frozen candidate employee name.
المؤهل الدراسي وتاريخه	Frozen qualification source and approved date presentation. Existing approved behavior of displaying qualification year only remains unless EGAS later approves a different visual date format.
الوظيفة الحالية	Frozen current job title.
الإدارة العامة	Frozen employee department/general administration from the annual employee data. Missing required value blocks final-form completion.
تاريخ شغلها	Frozen current-job start date / beginning of occupying the current job.
مدة الخبرة - سنة / شهر / يوم	Frozen annual-snapshot experience duration values as of the workbook-defined reference date.
التوصية - ترشيح / تأجيل	AUTH-controlled candidate recommendation field using the approved form values.
الوظيفة المرشح لشغلها	Effective nominated job. SAME_POSITION uses the current position. OTHER_POSITION uses a different approved position within the same department/routing unit and activates P40.
آخر تقرير كفاية مستوى	Frozen approved annual performance/report level; this returns to the official Promotion form under the new template.
ملاحظات تملأ بمعرفة النائب المختص	AUTH/deputy-controlled candidate notes for the official form, separate from generic internal audit metadata.
Signature area	Three official signer blocks: HR manager, ORG manager, AUTH manager/deputy.

9.4 Secondment official form baseline

- Business data and layout rules from the currently approved Secondment form remain applicable where not contradicted by v5.
- Hierarchy/signature behavior is updated to three business signers: HR manager at S1, ORG manager at S2 and AUTH manager/deputy at S3.
- S4/S5 are review/finalization stages and do not create duplicate ORG/HR signatures.
- Any future stakeholder-supplied revised Secondment form image may update the visual template/column mapping without changing the manager/employee workflow architecture unless it introduces new business rules.

10. Annual XLSX data requirements

10.1 Data-source policy

- The approved annual XLSX remains the authoritative operational employee snapshot.
- Importer validates required headers by normalized header name rather than fragile column position.
- No staff/authority workbook automatically creates application accounts or authorization. Admin controls accounts/hierarchy separately.
- Activation remains explicit and transactional. Failed validation/revalidation cannot silently replace the active snapshot.

10.2 New/retained normalized employee fields

Source concept	Normalized target / rule
رقم الموظف / رقم العامل	personnel_number - required unique business identifier.
اسم الموظف	employee_name - preserve Arabic text.
الوظيفة	current_job_title - frozen into request candidate.
الإدارة / الإدارة العامة	general_administration / department_name - required for the new official Promotion form.
النيابة / المساعد / routing source	source_routing_label -> explicit routing_unit_id mapping. Unmapped required values block workflow eligibility.
تقرير الكفاءة / performance	performance_rating / latest_report_level - frozen annual value and printed where required by the new official form.
المؤهل الأصلي / qualification fields	qualification source values and nullable source date. Existing form presentation may display year only.
تاريخ بداية الخبرة	experience_start_date - preserved source date.
عدد سنوات/شهور/أيام الخبرة حتى [reference date]	experience_years, experience_months, experience_days + experience_as_of_date.
بداية شغل الوظيفة	current_job_start_date.
عدد السنوات/الشهور/الأيام حتى [reference date] المرتبطة بشغل الوظيفة	job_tenure_years, job_tenure_months, job_tenure_days + job_tenure_as_of_date.

10.3 Duration handling rule

Authoritative annual values

When the approved workbook supplies both start dates and pre-calculated year/month/day durations, v5 stores both. The workbook-provided duration values are authoritative for the annual snapshot and official form. The importer may calculate a consistency warning, but it must not silently overwrite approved HR values with its own calculation.

- Reference dates embedded in headers (for example “1/1 حتى /YYYY”) are parsed/stored as snapshot metadata rather than encoded into database column names.
- A new annual workbook with a different reference year must not require a schema migration merely because the as-of date changed.
- Blank required new official-form data may be staged but must block that employee from final workflow completion/eligibility according to validation policy; the importer must report the missing value clearly.

11. PostgreSQL logical data model

11.1 Core entities

Entity	Purpose
UserAccount	Local login identity, password hash, status/lockout and profile metadata.
OperationalUnit	Hierarchy unit: HR, ORG or AUTHORITY_ROUTING. AUTH unit links to one routing authority/routing unit.
UserUnitMembership	One active operational unit membership for an operational account; level EMPLOYEE or MANAGER-capable through active manager assignment. Historical memberships preserved.
UnitManagerAssignment	Effective-dated manager/deputy assignment. Enforces one active manager per unit and preserves replacement history.
Employee / EmployeeAnnualSnapshot	Stable employee identity + immutable annual HR snapshot including new department/experience/job-tenure fields.
RoutingUnit / RoutingUnitSourceAlias	Approved routing values and explicit source-label mappings.
WorkflowRequest	Request header, type, current iteration/stage/status, originating routing context and final state.
WorkflowIteration	Immutable whole-request pass. Reject restart creates the next iteration.
RequestCandidate	Frozen candidate evidence/reference for the request iteration and official-form data.
StageExecution	Immutable execution/attempt of a business stage. Supports controlled inter-stage correction without rewriting old signed data.
WorkAssignment	Current/prior manager assignment to employee or manager self-work; reassignment history preserved.
WorkflowNote	Append-only request/candidate operational messages with author/unit/stage/execution attribution.
PromotionDecision	Same/Other Position, effective nominated job, recommendation and AUTH-controlled fields. Target routing must equal request/candidate routing. Any OTHER_POSITION decision activates conditional P4O for the whole Promotion request.
SecondmentPositionOption / SecondmentDecision	Organization-proposed options and AUTH-selected/approved position, restricted to same routing unit.
WorkflowSignoff	Immutable HR/ORG/AUTH signer evidence with manager assignment snapshot and signature hash. Password is never stored.
StageAction / AuditEvent / SecurityEvent	Append-only business actions, hierarchy/assignment events and security evidence.
Notification	Recipient-owned in-app notification/read state.
FinalDocument / PdfGenerationLog	Frozen final source/template/checksum and controlled generation audit.

11.2 Key integrity constraints

- Exactly one active manager assignment per operational unit.
- One operational account has at most one active operational-unit membership. Admin accounts cannot acquire workflow authority through the Admin role alone.

- Only one current WorkAssignment per active StageExecution; reassignment closes prior assignment and creates a new row/evidence atomically.
- Only the current accepted StageExecution may advance the business workflow.
- A signoff is unique for the relevant StageExecution/signature role and immutable after creation.
- Promotion/Secondment target routing-unit IDs must equal the request/candidate routing unit under v5 current business rules.
- Reject restart creates a new WorkflowIteration; it never resets/deletes prior stage execution/action/signoff rows.
- Completed final document source and checksum are immutable.

11.3 Migration baseline

001_initial_v5_schema.sql

002_seed_reference_routing_configuration.sql (only if approved static reference data is appropriate)

003+ future additive migrations only after v5 baseline is in use

Migration rule

Once a v5 migration is applied to a shared/pilot database, do not rewrite it. The clean-repository decision only means the old local v4 migration history is not imported as mandatory v5 history.

12. Backend modules and API principles

12.1 Modules

Module	Responsibility
auth	Login, sessions, password change/lockout, CSRF/Origin integration and current-account context.
admin	Account lifecycle, operational units, membership, manager replacement, routing/authority coverage and readiness.
hierarchy	Resolve manager/subordinates, authorize assignment/reassignment and historical manager context.
employee-data	Annual XLSX staging/validation/activation and employee lookup.
workflow	Request creation, iterations, StageExecution, WorkAssignment, manager/employee actions, correction/reject/restart/cancel.
promotion	P1-P5 stage-specific fields/decisions and validation.
secondment	S1-S5 stage-specific position/decision validation.
signatures	Signature asset validation/storage and password-reauthenticated immutable signoff.
documents	Current preview and final official PDF V3; separate audit PDF.
notifications	In-app actionable assignment/manager inbox notifications.
audit	Append-only business/security reporting.

12.2 API design rules

- Expose business commands, not generic table CRUD.
- Server derives actor account, manager/unit context, assignee and workflow state; client never supplies trusted actor IDs/roles.
- Every request/candidate/stage/assignment/document action is re-authorized on the server to prevent IDOR/BOLA.
- Use explicit allow-listed DTO validation and parameterized SQL.
- Stage transitions, assignment/reassignment, signoff and rejection/restart operations use one PostgreSQL transaction where atomicity is required.
- Bound and paginate history, notifications and Admin/audit lists.

12.3 Representative commands

Command / endpoint concept	Authorized actor / effect
POST /workflow/requests	HR manager/assigned HR preparation context; create draft P1/S1 request.
POST /workflow/stages/:id/assign	Current unit manager assigns/reassigns a subordinate or self.
POST /workflow/stages/:id/submit-to-manager	Current employee submits stage work to manager review.
POST /workflow/stages/:id/internal-correction	Current unit manager returns to current employee with reason.
POST /workflow/stages/:id/return-previous	Current unit manager returns to previous business-stage

	manager with reason.
POST /workflow/stages/:id/reject	Permitted current unit manager rejects; system routes control to HR manager.
POST /workflow/requests/:id/restart	HR manager only after rejection; creates new iteration.
POST /workflow/requests/:id/cancel	HR manager only after rejection; final cancellation.
POST /workflow/stages/:id/sign-and-advance	Required signing-stage manager/deputy + password; creates immutable signoff and advances atomically.
POST /workflow/stages/:id/approve-and-advance	Manager on non-signing review stage; advances without new official signature.

13. Security baseline

Control	v5 requirement
Password storage	Argon2id adaptive hashes only; no plaintext/reversible encryption/raw SHA password storage.
Sessions	Opaque random server-side sessions; HttpOnly cookies; configurable idle/absolute lifetime; account/password/hierarchy security changes revoke/rotate as required.
CSRF / Origin	Cookie-authenticated mutations require CSRF validation and trusted Origin checks.
Authorization	Deny by default. Backend checks account type, operational unit, manager assignment/current WorkAssignment, business stage and request scope. Frontend visibility is never the security boundary.
Hierarchy separation	Admin does not equal operational workflow authority. Operational user cannot act outside its single active unit. Employees cannot impersonate manager actions.
Signoff reauthentication	Every actual HR/ORG/AUTH manager signature requires the signer current raw password immediately before signoff insertion. Wrong/missing password creates no signoff.
Signature assets	Approved PNG/JPEG input only; decode/canonicalize, strip metadata, enforce byte/pixel limits, checksum and private randomized storage.
Concurrency	Transactional assignment/reassignment, stage advancement and compare-and-set/locking where needed. No process-only locks for correctness.
Audit/security logs	Append-only/tamper-evident business audit plus separate security events. Passwords/session/CSRF secrets never logged.
PDF boundary	No arbitrary user HTML/network/file:// input; controlled assets, bounded renderer queue/time/output and checksum-verified final document.
Deployment	Production HTTPS/Secure cookies, secrets outside Git, least-privilege runtime DB role and private storage.

13.1 Manager replacement security

- Replacing a manager is an Admin-controlled privileged action and is audited.
- A manager replacement affects future authorization only; historical signoffs remain tied to the old manager assignment snapshot.
- The system must prevent deleting/deactivating the last active manager assignment for a unit when active workflow tasks require a manager, unless a replacement is atomically installed.

14. Non-functional and UI requirements

14.1 Performance / reliability

ID	Requirement
NFR-PERF-01	Personnel Number lookup should target $\leq 1s$ p95 on EGAS LAN under normal load.
NFR-PERF-02	Normal workflow read/write/assignment actions should target $\leq 1.5s$ p95 excluding PDF/report work.
NFR-PERF-03	Server-side pagination/filtering required for request history, assignments, notifications and audit/report lists.
NFR-REL-01	Stage transitions, reassignment, signoff and evidence writes are transactional.
NFR-REL-02	PDF generation failure does not mutate workflow state.
NFR-REL-03	Historical iterations/stage executions/signoffs/final documents remain readable after later data/template/account changes.

14.2 UI/UX baseline

- Arabic is the default language; RTL tables/forms/navigation must handle long Arabic names and mixed IDs/numbers safely.
- Manager home screen centers on Manager Inbox, pending reviews, current assignments, rejected requests awaiting restart/cancel and employee workload/assignment actions.
- Employee home screen shows only currently assigned work plus authorized history/notifications.
- Assignment UI displays the previous worker as default when a request returns to a unit, while allowing the manager to choose another subordinate.
- Every stage clearly distinguishes إعادة للتصحيح, إعادة للمرحلة, internal حفظ, إرسال للمدير, manager sign/approve and final completion actions according to authorization.
- Current-form PDF preview is available to authorized workflow users; final completed request provides immutable final PDF and print/download.
- Target current company-approved Chrome/Firefox/Edge on laptop/desktop; WCAG 2.2 AA engineering goals remain appropriate.

14.3 Notifications / history

- In-app only in v5 unless separately approved.
- Incoming business stage notifies the responsible manager.
- Assignment/reassignment notifies the new assignee; former assignee loses edit access and may receive reassignment history notification if useful.
- Employee submission notifies manager; internal correction notifies employee; return-to-previous notifies destination manager; rejection notifies HR manager.
- Timeline displays manager/employee assignment events, stage submissions, corrections, returns, rejects, restart/cancel, signatures and completion according to authorization.

15. Testing and verification requirements

15.1 Hierarchy and authorization regression

- Exactly one active manager per operational unit; concurrent manager replacement cannot produce two active managers.
- Operational account cannot hold simultaneous active memberships that create multi-role workflow authority.
- Admin cannot perform operational workflow actions through Admin privileges.
- Only current manager and current assignee can edit an active stage.
- Former assignee loses edit access immediately after reassignment; historical actions remain visible.
- Manager can self-assign/take work and can reassign only to active subordinates of the same unit.
- AUTH task routes to the AUTH unit tied to the request routing unit.

15.2 Workflow regression

- Promotion SAME_POSITION path P1 -> P2 -> P3 -> P4 -> P5 with HR/ORG/AUTH manager signatures at P1/P2/P4.
- Promotion OTHER_POSITION path P1 -> P2 -> P3 -> P4 -> P4O -> P5 when any candidate is moved to a different position within the same department/routing unit; P4O is Organization confirmation and has no second ORG signature.
- Secondment path S1 -> S2 -> S3 -> S4 -> S5 with HR/ORG/AUTH manager signatures at S1/S2/S3 and no duplicate S4/S5 signatures.
- Employee submission always goes to manager, never directly to next business stage.
- Internal correction loops manager <-> current employee without changing business stage.
- Return for Correction moves exactly one business stage backward and preserves old StageExecution/signoff evidence.
- Returning to a signed stage requires a new StageExecution and fresh manager signature before re-advancement.
- Reject routes to HR manager; HR manager can restart whole request as next iteration or cancel.
- Restart preserves all previous iteration evidence.
- Promotion OTHER_POSITION and Secondment selected targets cannot cross routing unit/department; backend rejects such targets. Promotion OTHER_POSITION remains a same-department position change and activates P4O.

15.3 Signature/security regression

- Correct current manager/deputy password succeeds for each actual signature.
- Missing/wrong/another-account password creates no signoff.
- Password is not trimmed/transformed before verification and is never persisted/logged/snapshotted.
- Only the active manager assignment can sign for that unit/stage. An employee or replaced manager cannot sign unless currently authorized.
- Duplicate signoff for the same StageExecution/role is rejected.

15.4 Data/PDF regression

- Required new XLSX headers/fields are detected by normalized header name and reference-date metadata is parsed without year-specific DB columns.
- Workbook-provided experience/job-tenure durations are preserved; any calculated consistency check does not silently overwrite them.
- Final Promotion PDF V3 contains the new official columns and three signer blocks.
- Final PDF cannot be frozen while mandatory fields or current accepted signatures are missing.
- Current preview can change before completion; final PDF source/checksum remains immutable after completion.
- Arabic RTL/form layout and long names/job titles are visually verified, not only structurally tested.

16. Acceptance criteria

- Admin can create operational units/accounts, attach employees to one unit, configure exactly one active manager per unit and replace that manager with preserved history.
- Incoming workflow stages are manager-controlled and can be assigned/reassigned to subordinate employees or worked by the manager directly.
- Only current manager/current assignee can edit active work; reassignment revokes the old assignee edit authority immediately.
- Promotion implements P1-HR -> P2-ORG -> P3-HR -> P4-AUTH -> P5-HR for all-SAME_POSITION decisions, and P1-HR -> P2-ORG -> P3-HR -> P4-AUTH -> P4O-ORG -> P5-HR when any candidate is OTHER_POSITION. HR/ORG/AUTH manager signatures occur only at P1/P2/P4; P4O does not duplicate the ORG signature.
- Secondment implements S1-HR -> S2-ORG -> S3-AUTH -> S4-ORG -> S5-HR and has HR/ORG/AUTH manager signatures only at S1/S2/S3.
- Promotion Same/Other Position and Secondment targets cannot leave the request department/routing unit. P4O remains active only for Promotion: any same-department OTHER_POSITION decision sends the whole request to P4O; all-SAME_POSITION decisions bypass it.
- Internal correction, return-to-previous-stage and complete rejection are implemented as three distinct commands with required reasons.
- Rejection always routes control to HR manager; only HR manager decides restart whole request or cancel.
- Restarting creates a new immutable iteration while retaining prior rejection/actions/signatures/documents.
- Every actual manager/deputy signature requires fresh current-password reauthentication and creates immutable signer evidence.
- Annual XLSX stores new department, experience and job-tenure fields with workbook-provided duration values treated as authoritative snapshot data.
- Final Promotion official PDF follows the supplied V3 column/signature contract and freezes exactly one immutable final official document at completion.
- All authorization is enforced server-side and tests include IDOR, reassignment, replaced-manager, concurrency and signoff cases.
- New repository starts from a clean v5 migration baseline; no obsolete v4 migration assumptions are required for a fresh installation.

17. Implementation phases for the new repository

Phase	Scope	Exit condition
0 - Archive / bootstrap	Tag/archive v4 repo; create v5 repository; establish CI, lint/test/build, environment and clean PostgreSQL migration runner.	New repo builds/tests; old repo remains preserved.
1 - Identity & hierarchy	Auth/session security; Admin accounts; operational units; user membership; one active manager; manager replacement; subordinate queries.	Hierarchy authorization and manager replacement regression complete.
2 - Annual data V5	Importer updates for department, experience, job-tenure/reference-date fields; routing mapping; activation.	Synthetic annual snapshot validates/activates and employee lookup exposes required fields.
3 - Workflow engine	WorkflowRequest/Iteration/StageExecution/WorkAssignment; manager inbox; assignment/reassignment; employee submit; manager review; corrections/reject/restart/cancel; notes/timeline/notifications.	Generic stage model proven without Promotion/Secondment-specific decisions.
4 - Promotion	P1-P5 plus conditional P4O; Same/Other Position same-routing constraint; OTHER_POSITION -> P4O Organization confirmation; recommendation/nominated job; HR/ORG/AUTH signatures.	Promotion end-to-end regression passes.
5 - Secondment	S1-S5 positions/decision/confirmation; same-routing constraint; HR/ORG/AUTH signatures.	Secondment end-to-end regression passes.
6 - PDF V3 / audit	Current preview, final Promotion V3, revised Secondment signers, final snapshot/checksum, audit PDF.	Visual Arabic PDF QA and immutability tests pass.
7 - Frontend completion	Manager/employee/Admin screens, assignment/review loops, new form fields and responsive RTL UX.	Stakeholder workflow UAT-ready UI.
8 - Hardening / UAT	Performance, browser/accessibility, backup/restore, cybersecurity review, genuine data/configuration, stakeholder UAT.	Pilot acceptance gates complete.

Implementation sequencing recommendation

Build the generic hierarchy + StageExecution/WorkAssignment engine before duplicating logic inside Promotion and Secondment. This keeps manager assignment/correction/rejection behavior shared and reduces coupling.

18. Architecture decision records

ADR	Decision
ADR-001	Create a new v5 repository/database baseline because all v4 implementation changes are local/disposable and the workflow/authorization model is materially redesigned.
ADR-002	Use a modular monolith: React/Vite SPA + Express 5/TypeScript + pg + PostgreSQL.
ADR-003	Represent department/authority hierarchy with OperationalUnit + effective manager assignment +

	subordinate membership.
ADR-004	Operational accounts have one active operational-unit membership; remove multi-role active-role switching from v5.
ADR-005	Separate BusinessStage from internal manager/employee WorkAssignment state.
ADR-006	Use immutable StageExecution attempts to support corrections to already signed stages without rewriting signoffs.
ADR-007	Reject always returns control to HR manager; restart creates a new WorkflowIteration or HR manager cancels.
ADR-008	HR/ORG/AUTH manager/deputy are the three official signers; later review stages do not duplicate prior signatures.
ADR-009	Every actual signature requires fresh current-password reauthentication; password is never stored.
ADR-010	Promotion/Secondment target positions remain within the same department/routing unit. Promotion P4O is retained as conditional Organization confirmation when any candidate is OTHER_POSITION; all-SAME_POSITION decisions bypass P4O.
ADR-011	Use one immutable final official PDF rather than immutable PDF per internal assignment/receive event; audit records preserve workflow evidence.
ADR-012	Workbook-provided annual duration values are authoritative; date arithmetic may warn but cannot silently replace them.
ADR-013	New Promotion official form uses a V3 template contract with department/tenure/experience/recommendation/nominated-job/report fields and three signer blocks.

19. Traceability to stakeholder decisions and prior baseline

Decision / evidence	v5 coverage
Stakeholder: HR/ORG/AUTH each use manager + subordinate employee hierarchy	Sections 4-8, 11, 15-18.
Stakeholder: manager assigns/reassigns, may work personally, employee returns work to manager	Sections 4-5.
Stakeholder: later return to same unit defaults to prior worker but manager may change	Section 4.4.
Stakeholder: only current assignee and manager have active work access	Section 4.3.
Stakeholder: reject returns to HR manager for restart/cancel; correction may go to previous stage or employee	Section 6.
Stakeholder: one manager per HR/ORG and one manager/deputy with employees per AUTH routing unit; Admin manages/replaces	Section 4.
Stakeholder: remove multi-role behavior if it complicates system	Sections 4.1 and ADR-004.
Stakeholder: HR/ORG/AUTH are official signers; later review stages do not duplicate signatures	Sections 7-9.
Stakeholder: password before each actual signature	Sections 9.2, 13 and 15.3.
Stakeholder: Promotion Same/Other and Secondment positions stay in same department; Promotion OTHER_POSITION still activates P4O	Sections 6-8; P4O retained with corrected same-department OTHER_POSITION trigger.
Supplied Promotion form image: new department/job-start/experience/recommendation/nominated-job/report/deputy-notes columns and third signer	Sections 9.2-9.3.
Supplied annual Excel header image: experience start, experience durations, current-job start and job-tenure duration fields	Section 10.
Prior v4 technical/security/data patterns still useful where non-conflicting	Sections 3, 10, 12-14.

19.1 Source documents / artifacts

- EGAS_Requirements_Architecture_Baseline_v4.0_Current(1).pdf - historical baseline and prior implementation-aligned requirements.
- Stakeholder-supplied Promotion official-form image (19 Aug 2026 discussion context).
- Stakeholder-supplied annual Excel header excerpt showing new experience/current-job tenure fields.
- Stakeholder clarifications captured in the project conversation on 19 Aug 2026.

20. Final implementation handoff

Current source rule

For the new repository, implement v5.2. Do not copy v4 workflow behavior merely because code already exists. Reuse old technical code only after checking that its authorization, schema and state transitions match this document. P4O is active only under the same-department OTHER_POSITION trigger defined here.

- Begin with the new repository/bootstrap and hierarchy schema; do not start by porting the old workflow tables wholesale.
- Treat the manager/employee stage model as shared infrastructure used by both workflows.
- Keep old repo read-only/tagged so any useful test/security/PDF implementation can be ported deliberately.
- Do not invent cross-routing Promotion/Secondment targets, multi-role operational accounts, extra signatures, actor-received immutable PDFs or alternative rejection semantics. P4O must remain conditional on same-department OTHER_POSITION Promotion decisions, not on cross-department routing.
- The revised Secondment visual template can be incorporated later if EGAS supplies a new official form; this does not block the hierarchy/workflow architecture defined here.

21. Detailed implementation design blueprint

This section is normative for implementation structure but does not change the stakeholder business behavior already frozen in Sections 1-20. It exists so implementation agents can work from explicit boundaries, states, entities and transaction rules instead of reverse-engineering them from historical code.

21.1 Historical repository review basis and reuse policy

The historical repository was reviewed as a technical reference before defining the v5.2 detailed design. Its Express/pg modular-monolith structure, database helpers, annual-data pipeline, workflow services, signature/PDF implementation and audit/notification patterns are useful references. Its flat/multi-role authorization and old workflow schema are not implementation authority for v5.2.

Historical reference	Observed pattern	v5.2 treatment
Express application boundary	<code>`services/api/src/app.ts`</code> mounts auth, admin, reference, employee-data, workflow and notifications behind common middleware.	Reuse the thin route/module mounting pattern.
Backend module layout	<code>`services/api/src/modules/`</code> contains admin, audit, auth, authorities, employee, health, import, notifications, reference and workflow.	Retain modular monolith; split v5 workflow-specific behavior into workflow-core, promotion, secondment, hierarchy, signatures and documents.
Database layer	<code>`services/api/src/db/`</code> contains pool, transaction, migration runner, repositories and immutable historical migrations.	Reuse connection/transaction/migration-runner patterns; start clean new schema at v5 migration 001.
Historical workflow implementation	workflow module contains promotion-service, secondment-service, signature-service, task-service, PDF/history/form-snapshot services.	Reuse tested technical patterns after business-rule review; do not copy old task ownership, P4O routing or signoff uniqueness blindly.
Historical schema	baseline/migrations model accounts, roles/sessions, annual snapshots, authority assignments, requests/iterations/tasks, signoffs, PDF and audit evidence.	Use as entity discovery reference only; v5 physical schema is redesigned around <code>OperationalUnit</code> , <code>UnitManagerAssignment</code> , <code>StageExecution</code> and <code>WorkAssignment</code> .
Migrations 001-007	Historical repository has migrations through <code>`007_promotion_cross_department_review.sql`</code> .	Do not import them into the new repository because all old implementations are local/disposable and v5 starts from a clean database baseline.

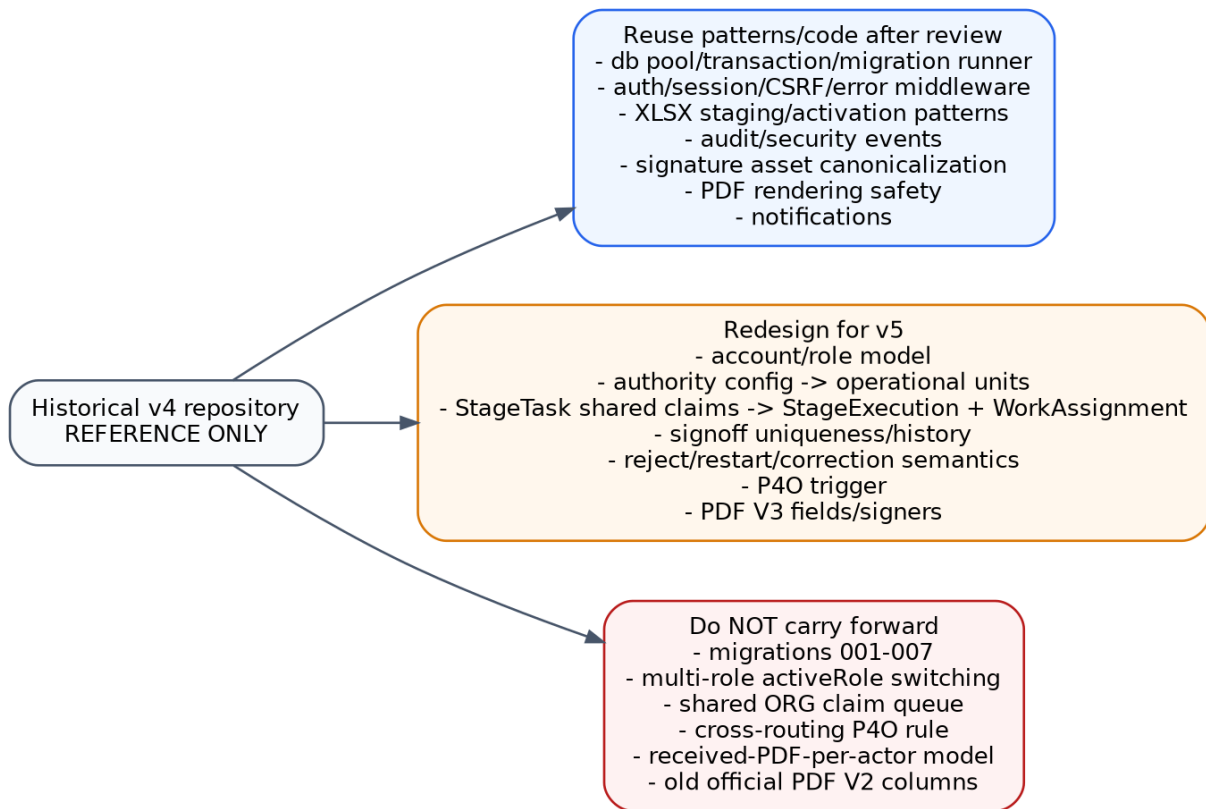


Figure 7 - Historical repository reuse / redesign / do-not-carry-forward map

21.2 Functional component architecture

The new repository remains a modular monolith. Domain modules may call each other only through explicit services; database access remains behind repositories/transaction helpers. Promotion and Secondment use the generic workflow engine rather than duplicating assignment/correction/reject logic.

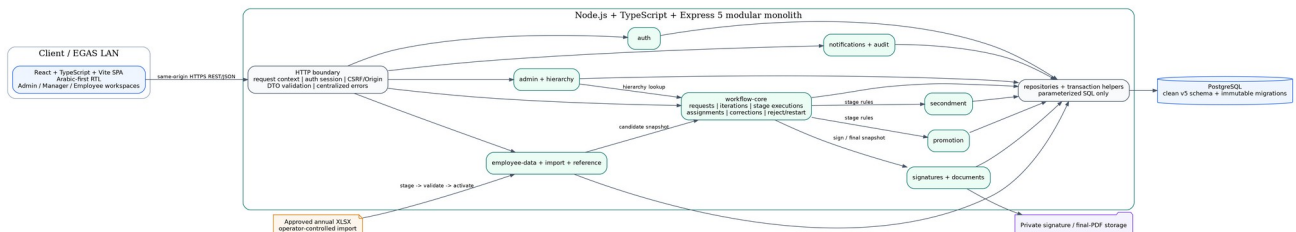


Figure 8 - v5.2 functional component architecture

Module	Owned responsibility
auth	Login, Argon2id verification, server-side sessions, password change/lockout, current-user context.
admin	Account lifecycle, routing/reference configuration, operational-unit administration and readiness.
hierarchy	Membership, current manager resolution, subordinate lists, manager replacement/history and assignment authorization.
employee-data / import	Annual XLSX stage/validate/activate, routing aliases, immutable employee snapshots and lookup.
workflow	Requests, iterations, StageExecution, WorkAssignment, manager/employee loop, notes, return/reject/restart/cancel and generic advancement.
promotion	P1-P5/P4O stage-owned fields, Same/Other Position

	decisions and P4 conditional branch.
secondment	S1-S5 position options, AUTH selection and S4 confirmation constraints.
signatures	Signature assets, manager password reauthentication, immutable signoff snapshot.
documents	Current preview, final form snapshot, PDF V3 materialization and audit PDF.
notifications / audit	Recipient-owned notifications, append-only business/admin/security evidence.

21.3 StageExecution internal state machine

A StageExecution represents one attempt/execution of one business stage. Manager/employee handoff is internal state, not a new P/S stage. An inter-stage correction creates a fresh StageExecution of the previous business stage; a whole-request restart creates a fresh WorkflowIteration.



Figure 9 - Generic StageExecution manager/employee work-state machine

21.4 Request lifecycle state machine

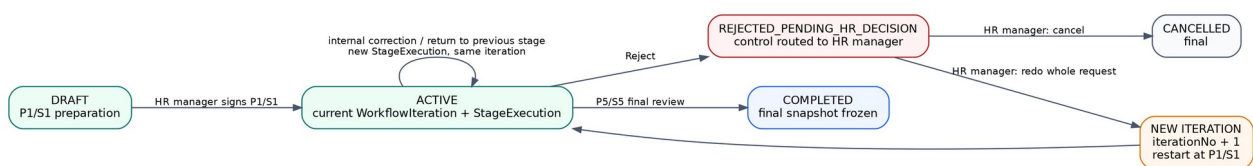


Figure 10 - WorkflowRequest / WorkflowIteration lifecycle including reject recovery

Request/iteration state	Meaning
DRAFT	Initial P1/S1 preparation before first HR-manager signoff.
ACTIVE	Current iteration is progressing through business stages.
REJECTED_PENDING_HR_DECISION	A manager rejected; current iteration is closed and HR manager alone chooses restart or cancel.
COMPLETED	P5/S5 final review accepted; final form snapshot is immutable.
CANCELLED	HR manager cancelled after reject; final request state.
Iteration restart	Creates WorkflowIteration N+1; never resets/deletes prior executions, signoffs or snapshots.

21.5 Server authorization decision flow

Every workflow object is authorized server-side from the authenticated account, effective membership/manager assignment and current WorkAssignment. Browser-supplied user IDs, manager flags or role text are never authorization inputs. Admin authority never implies operational workflow authority.

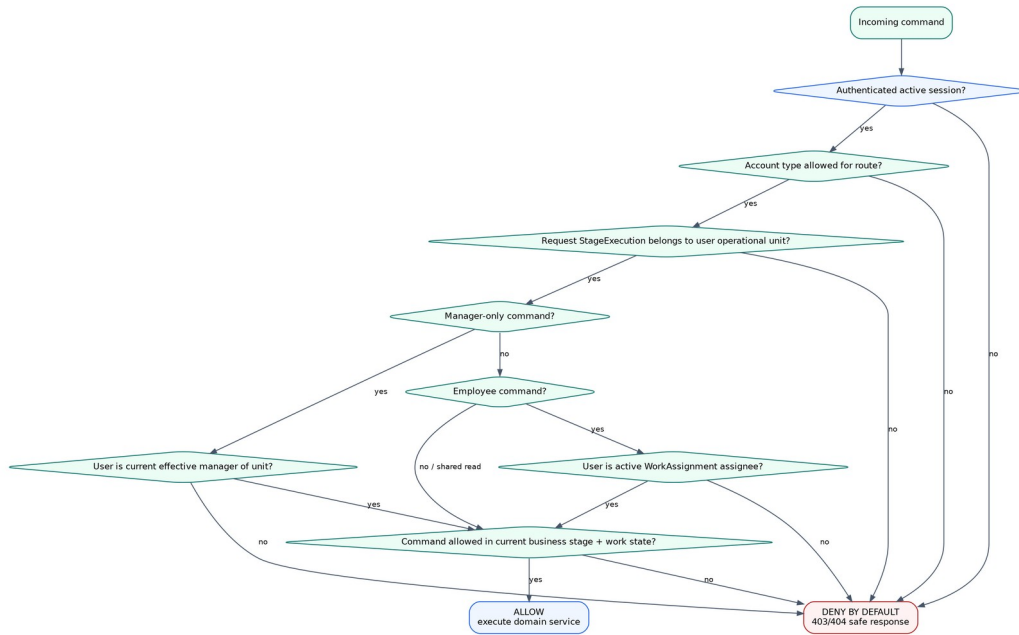


Figure 11 - Deny-by-default workflow authorization decision flow

22. Database physical design contract and ERDs

The v5 repository starts with a clean PostgreSQL migration 001. The ERDs below are the canonical logical/physical direction for that migration. Lowercase snake_case table names are recommended for the new schema; an implementation agent must not recreate the historical `egas\_\*` CamelCase/flat-role schema merely for parity with v4.

22.1 Identity and hierarchy ERD

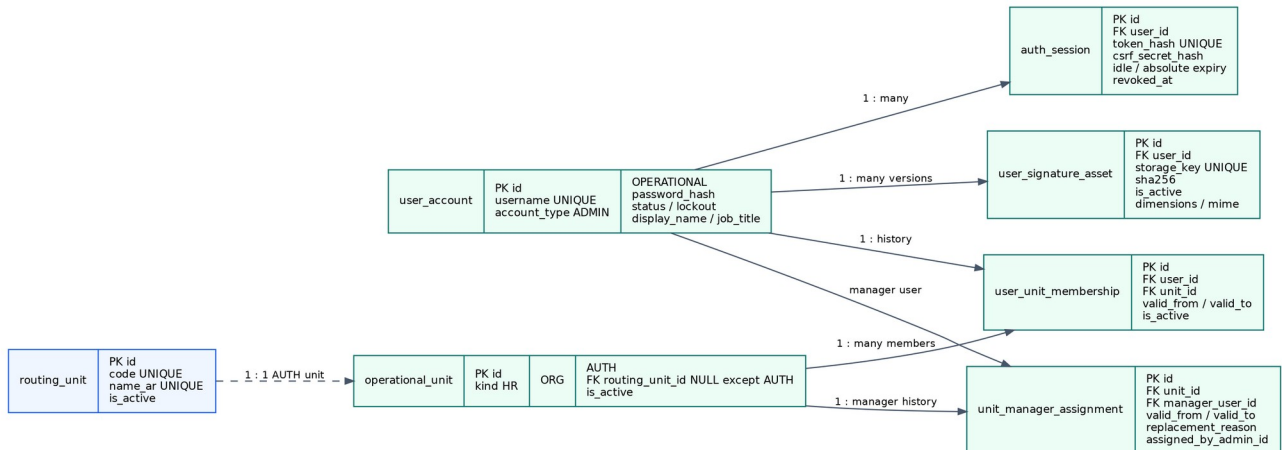


Figure 12 - Identity, operational-unit membership, manager history and signature assets

- `user\_account.account\_type` is ADMIN or OPERATIONAL. Admin accounts do not gain workflow authority.
- An operational account has one active UserUnitMembership. Manager authority comes from an effective UnitManagerAssignment, not job-title text.
- There is one active HR unit, one active ORG unit and one active AUTH operational unit for each configured RoutingUnit. Each unit has one active manager at a time.

22.2 Annual employee/import ERD

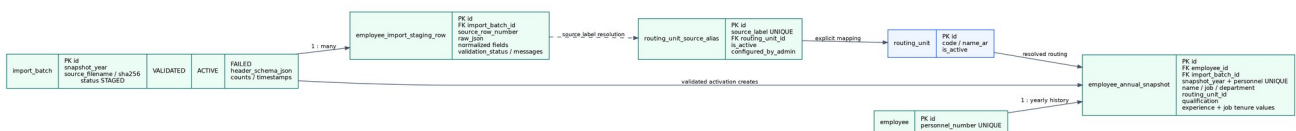


Figure 13 - Approved XLSX staging, validation, routing resolution and immutable annual snapshots

- The annual workbook remains authoritative for provided experience/job-tenure duration values; the app stores source dates and source duration values rather than silently recalculating authoritative HR values.
- Unmapped non-null routing labels block affected rows until an explicit RoutingUnitSourceAlias exists.

22.3 Workflow core ERD

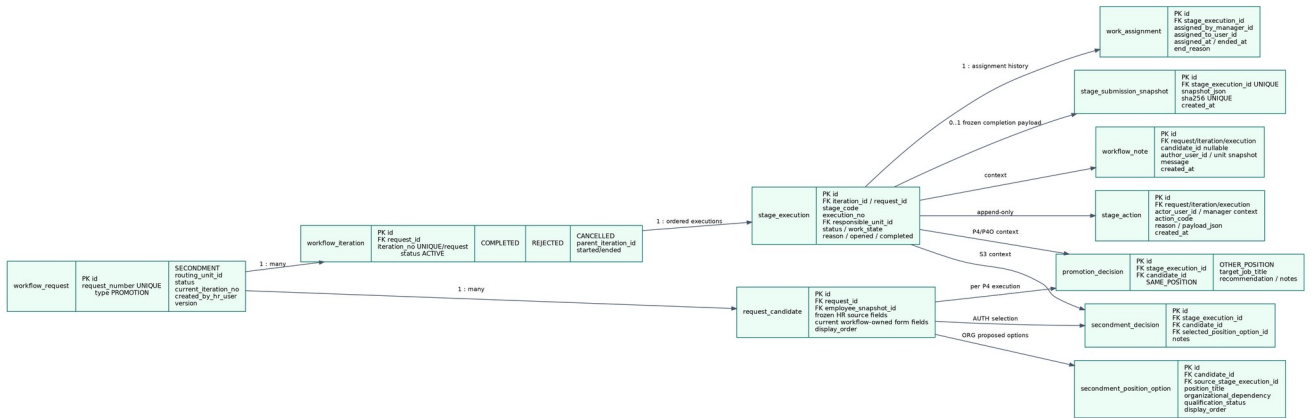


Figure 14 - Workflow request, iteration, stage execution, assignment and workflow-specific data

Key redesign: a StageExecution is an immutable stage attempt boundary. WorkAssignment stores assignment/reassignment history. StageSubmissionSnapshot freezes what the manager accepted/signed at completion. Returning to an already completed stage creates a new StageExecution and therefore a new signoff when that stage is a signing stage; it never overwrites the previous execution/signoff.

22.4 Documents, signoff, notifications and audit ERD

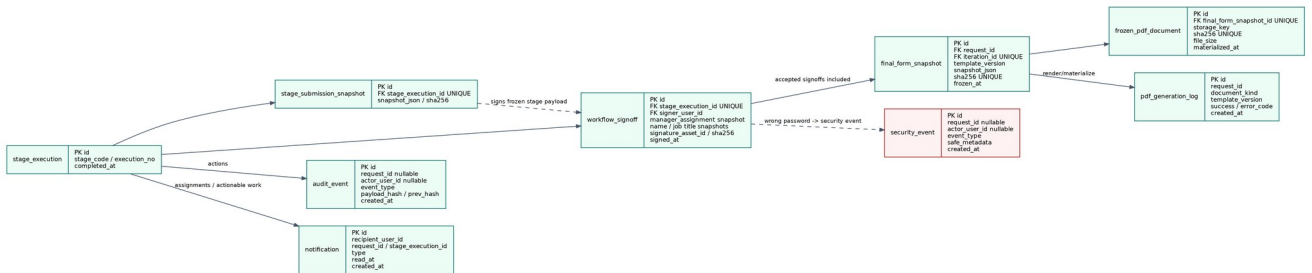


Figure 15 - Stage snapshots, signoffs, final document evidence, notifications and audit/security events

22.5 Canonical physical tables and required constraints

Canonical table	Minimum contract / invariant
user_account	username unique; account_type ADMIN/OPERATIONAL; Argon2id password hash; active/lockout/profile metadata.
auth_session	opaque token hash unique; user FK; CSRF secret hash; idle/absolute expiry; revocation metadata.
routing_unit	business department/niaba routing values; code/name unique; active flag.
operational_unit	kind HR/ORG/AUTH; AUTH requires routing_unit_id; one active HR, one active ORG, one active AUTH unit per routing unit.
user_unit_membership	effective membership history; one active membership per operational user.
unit_manager_assignment	effective manager/deputy history; one active manager per unit; manager must be active member of same unit.
user_signature_asset	private canonicalized image metadata + SHA-256 + active/version history.
import_batch	annual workbook source/hash, detected headers, validation/activation status and row counts.
employee_import_staging_row	raw row JSON + normalized columns + routing resolution + validation messages.
employee	stable personnel_number unique identity.
employee_annual_snapshot	immutable yearly source values; unique(snapshot_year,

	personnel_number); includes department, qualification, performance/report, experience and job-tenure values.
routing_unit_source_alias	explicit source label -> routing_unit; source label unique.
job_category_reference	approved form grouping/category reference.
qualification_status_reference	approved Secondment qualification/compliance values.
workflow_request	request number/type/routing/status/current iteration/stage/version; completion/cancel timestamps.
workflow_iteration	request + iteration_no unique; immutable pass; parent/restart/reject metadata.
request_form_section	request/category/display order; request/category unique.
request_candidate	request + employee snapshot unique; frozen source evidence plus current accepted business fields.
stage_execution	iteration/stage/execution_no unique; responsible unit; status/work_state; open/complete/return metadata.
work_assignment	assignment history; manager -> assignee; one active assignment per StageExecution.
stage_submission_snapshot	stage_execution unique; immutable accepted/submitted JSON + SHA-256.
workflow_note	append-only request/candidate note; author/user/unit/iteration/stage-execution attribution.
stage_action	append-only domain actions with actor context, reason and safe payload.
promotion_decision	one per candidate per P4 execution; SAME_POSITION/OTHER_POSITION; OTHER requires target_job_title; same routing only; recommendation/notes.
secondment_position_option	ORG-proposed option tied to candidate + source S2 execution; title/dependency/qualification status/order.
secondment_decision	one selected valid option per candidate per S3 execution.
workflow_signoff	stage_execution unique; signer + unit/manager/name/job/signature snapshots; signed_at; never stores password.
final_form_snapshot	one immutable completed-iteration source; template version + JSON + SHA-256.
frozen_pdf_document	materialized final PDF storage key/hash/size; tied to final form snapshot.
pdf_generation_log	controlled render attempts/result without changing workflow state.
notification	recipient-owned request/execution notification + read state.
audit_event	append-only business/admin evidence; tamper-evident chain/hash where implemented.
security_event	append-only safe security evidence; no password/session/CSRF secret content.

22.6 Critical uniqueness and history rules

Invariant	Required behavior
Active membership	At most one active UserUnitMembership per OPERATIONAL user.
Active manager	At most one active UnitManagerAssignment per OperationalUnit; manager is a member of that unit.
AUTH mapping	One active AUTH OperationalUnit per active RoutingUnit.

Current assignee	At most one active WorkAssignment per StageExecution.
Stage execution identity	Unique (iteration_id, stage_code, execution_no). Return-to-previous creates a higher execution_no; do not reopen old executions.
Stage snapshot	Exactly one immutable StageSubmissionSnapshot when an execution completes.
Signoff	At most one WorkflowSignoff per StageExecution. Re-sign after correction occurs on a new execution, not by updating the old signoff.
Promotion decisions	Unique current decision per candidate per P4 execution. Any OTHER_POSITION routes whole request to P4O.
Secondment decision	Exactly one valid ORG-proposed option selected per candidate in each accepted S3 execution.
Final evidence	FinalFormSnapshot is immutable for completed iteration; PDF materialization may retry but cannot mutate workflow state.

22.7 Transaction and locking boundaries

Atomic command	Required transaction boundary
Assign / reassign	Lock StageExecution; verify manager + target membership; end active WorkAssignment; insert new assignment; action/audit/notification; commit.
Employee submit to manager	Lock StageExecution + active assignment; verify assignee; validate owned fields; set MANAGER_REVIEW; evidence + manager notification; commit.
Internal correction	Lock execution; verify manager; reason required; set CORRECTION_REQUIRED; preserve/reassign explicitly; evidence + employee notification; commit.
Return previous	Lock request/iteration/current execution; close current; create new previous-stage StageExecution with next execution_no; route to previous manager inbox; evidence + notification; commit.
Reject	Lock request/iteration/execution; close iteration REJECTED; request -> REJECTED_PENDING_HR_DECISION; create HR manager recovery task/control; evidence + notification; commit.
Restart / cancel	HR manager only. Restart creates WorkflowIteration N+1 + P1/S1; cancel sets final CANCELLED; preserve prior evidence.
Sign and advance	Lock account + StageExecution; re-check manager/state; Argon2id verify; freeze stage snapshot; insert signoff; complete/create next execution; audit/notifications; commit.

23. Runtime sequence designs

23.1 Password-reauthenticated sign and advance

Signing is a domain command, not generic signoff CRUD. The same transaction that creates the immutable signoff also completes the current StageExecution and creates the next execution. Wrong/missing password creates no signoff and no stage transition.



Figure 16 - Manager password reauthentication, immutable signoff and atomic advancement

Rule	Implementation contract
Promotion signing stages	P1 HR manager; P2 ORG manager; P4 AUTH manager/deputy.
Secondment signing stages	S1 HR manager; S2 ORG manager; S3 AUTH manager/deputy.
No duplicate later signatures	P3/P40/P5 and S4/S5 are review/confirmation stages; existing HR/ORG signatures remain the official signer evidence.
Password handling	Raw current signer password is verified using Argon2id immediately before signoff; never trim, log, snapshot or persist it.
Signer evidence	Freeze current manager assignment, user identity/name/job title, signature asset ID/hash and server signed_at.
Correction to signed stage	Create a new StageExecution and new signoff; prior signoff remains immutable historical evidence.

23.2 Promotion P4 conditional P40 sequence



Figure 17 - Promotion AUTH decision/signoff and conditional same-department P40 branch

- P4 decisions are per candidate. OTHER_POSITION means a different job/position in the same request department/routing unit; cross-routing is invalid.
- If any accepted P4 decision is OTHER_POSITION, create one P40 StageExecution for the whole request.
- P40 Organization can confirm, internally correct its employee, return to P4 AUTH or reject to HR manager recovery. It cannot mutate the signed P4 decision and does not create a second ORG signature.

23.3 Reject -> HR manager -> restart/cancel sequence

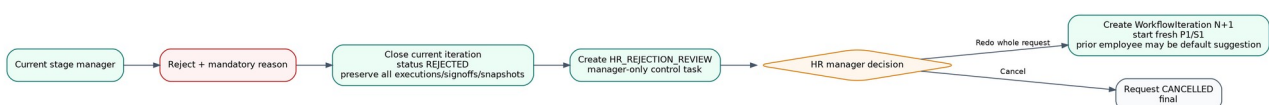


Figure 18 - Complete rejection recovery and immutable whole-request restart

24. Deployment and security trust-boundary design

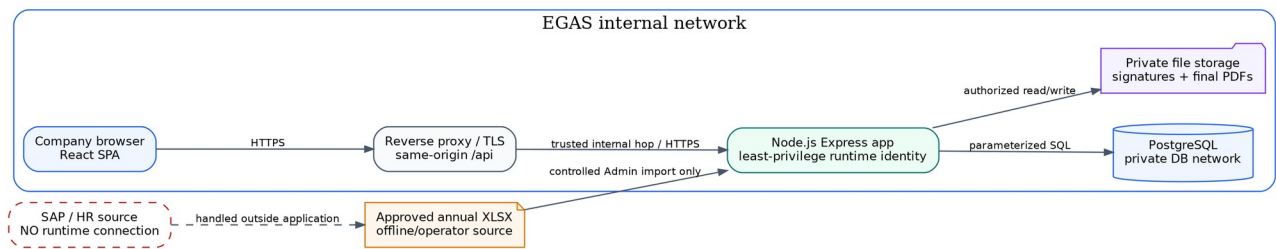


Figure 19 - EGAS internal deployment topology and trust boundaries

The application is standalone and internally hosted. The approved annual XLSX is the only operational employee-data ingress in v5. Normal browser operation has no SAP database/API path and no direct browser-to-PostgreSQL connection. Production uses same-origin HTTPS, a restricted application DB identity and private signature/PDF storage.

24.1 Security-boundary implementation rules

Boundary	Required control
Browser -> reverse proxy	HTTPS only in production; same-origin API; Secure/HttpOnly cookies; no wildcard credentialed CORS.
Reverse proxy -> API	Preserve trusted Origin/Host policy; request size/time limits; security headers.
API -> PostgreSQL	Least-privilege runtime role; parameterized SQL; migration/DDDL owner separate.
API -> private storage	Randomized storage keys; authorization on reads; canonicalized signatures; checksum-verified final PDFs.
XLSX import	Admin/operator-controlled only; header validation, staging, explicit activation; raw file/data never committed to Git.
PDF renderer	No arbitrary HTML/network/file:// input; controlled assets; bounded queue/time/output; PDF failure cannot change workflow state.
Audit/security	Append-only; passwords/session/CSRF secrets excluded from all metadata/logs.

25. SDLC and implementation-agent dependency map

The purpose of this section is to minimize implementation-agent context and rework. Agents should receive only the current phase contract plus stable dependency interfaces, not the full historical chat/project narrative.

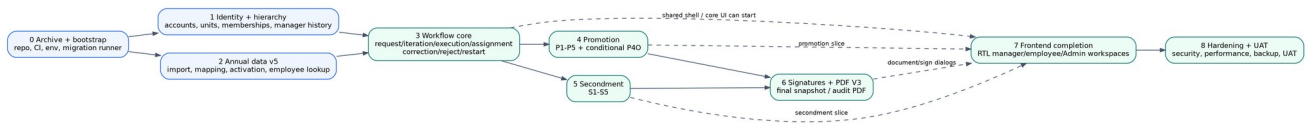


Figure 20 - Recommended implementation phase dependency and frontend vertical-slice timing

25.1 Agent work packages

Work package	Read first	Expected output	Exit gate
0 - Bootstrap	Sections 3, 11.3, 12, 13, 24-25	Repo skeleton, CI, env example, migration runner, DB bootstrap, build/test/typecheck/security scripts.	No business workflow implementation.
1 - Identity & hierarchy	Sections 4, 13, 21.5, 22.1, 22.5-22.7	Accounts/sessions, operational units/memberships, manager replacement, hierarchy APIs/authz tests.	One active manager/unit; one operational membership/user; Admin no workflow authority.
2 - Annual data v5	Sections 10, 22.2, 24	XLSX v5 stage/validate/activate, routing aliases, employee snapshots/lookup.	Synthetic import passes; authoritative values preserved; unresolved routing blocks rows.
3 - Workflow core	Sections 5-6, 11-12, 21.3-21.5, 22.3, 22.5-22.7	Request/iteration/execution/assignment engine, notes/timeline, correction/reject/restart/cancel, notifications.	Concurrency + IDOR + history tests pass before Promotion/Secondment logic.
4 - Promotion	Section 7, 9, 15, 23.1-23.2	P1-P5/P40 fields, decisions, sign/advance integration, end-to-end tests.	Any OTHER_POSITION -> P40; same routing only; three accepted signer blocks available to final form.
5 - Secondment	Section 8, 9, 15, 23.1	S1-S5 options/selection/confirmation, sign/advance integration, end-to-end tests.	Exactly one valid AUTH selection/candidate; same routing only; S4 no replacement of signed S3 result.
6 - Documents	Sections 9, 13, 22.4, 23.1, 24	Current preview, FinalFormSnapshot, Promotion V3 + revised Secondment PDF, audit PDF, checksum/render logs.	Arabic visual QA; immutable final evidence; PDF failure never mutates workflow state.
7 - Frontend	Sections 14, 21-23 + stable API docs	Admin, manager inbox, employee work, assignment/review loops, Promotion/Secondment forms, sign dialog, PDF/history.	RTL UAT-ready UI; frontend never treated as authorization boundary.
8 - Hardening/UAT	Sections 13-16, 24	Performance/browser/allly, backup/restore, security review, genuine config/data, stakeholder UAT.	Pilot acceptance gates complete.

25.2 Coding-agent source precedence

Priority	Source	Rule
1	v5.2 Requirements & System Architecture Baseline	Business, workflow, security, data and architecture authority.
2	EGAS Implementation Blueprint v1.0	Machine-readable naming, module, entity, transaction and work-package map. Must not override baseline business rules.
3	New repository applied migrations / current API docs	Physical schema and implemented interface authority after each phase is accepted.

4	Current new-repository code/tests	Implementation evidence; discrepancies must be treated as defects or approved changes, not silently reinterpreted.
5	Historical v4 repository	Reference for reusable technical patterns only; never source current v5 workflow behavior from old code alone.

Agent context rule: an implementation prompt should cite the exact baseline sections, blueprint sections, dependency API docs, current migration(s) and test expectations for that work package. Do not provide the entire old project history unless investigating a specific reusable implementation detail.

Implementation-ready source rule

The v5.2 baseline and its companion implementation blueprint define the target system. Historical code is a reference, not a specification. Implement the generic hierarchy + StageExecution/WorkAssignment engine once, then layer Promotion and Secondment stage rules on top. Preserve immutable execution/signoff/audit history and keep all authorization server-side.